Conventional and Unconventional Machining Processes

Machining is the process of removing unwanted material from a workpiece to obtain the required shape, size, and surface finish.

There are **two main types**:

1. Conventional Machining Process
2. Unconventional (Non-Traditional) Machining Process

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1. Conventional Machining Process

Definition

Conventional machining removes material by **direct contact** between a cutting tool and the workpiece. The cutting tool is harder than the material being machined.

Principle

Material removal occurs due to **mechanical cutting**.

Examples

- Turning
- Drilling
- Milling
- Shaping
- Planing
- Grinding
- Sawing
- Broaching

A) Turning (Lathe Machine)

Diagram

```
 Cutting Tool
      ↓
------------------------> Feed
========================  Workpiece
        ↻ Rotating
```

Explanation

- Workpiece rotates.
- Cutting tool moves linearly.
- Produces cylindrical components.

Examples

- Shafts
- Bolts
- Bushes
- Pulleys

B) Drilling

Diagram

```
 Drill Bit
    ↓
    |
    |
____________
 Workpiece
```

Explanation

- Rotating drill makes a circular hole.

Examples

- Bolt holes
- Pipe holes

C) Milling

Diagram

```
Rotating Cutter
      ↻
  0000000
______________
 Workpiece →
```

Explanation

- Rotating cutter removes material.
- Workpiece moves against the cutter.

Examples

- Gear teeth
- Slots
- Keyways

D) Grinding

Diagram

```
Grinding Wheel
      ↻
0000000000
-----------
Workpiece →
```

Explanation

- Abrasive wheel removes a very small amount of material.
- Used for high accuracy and smooth finish.

Examples

- Bearing surfaces
- Cutting tools

Advantages of Conventional Machining

- Simple machines
- Low operating cost
- High material removal rate
- Easy maintenance
- Suitable for most metals

Disadvantages

- Tool wear is high.
- Cannot easily machine very hard materials.
- High cutting forces.
- Limited for intricate shapes.

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2. Unconventional (Non-Traditional) Machining Process

Definition

Material is removed **without direct mechanical contact** or with very little contact, using electrical, chemical, thermal, or high-energy methods.

Principle

Material removal occurs by:

- Electrical discharge
- Chemical reaction
- Laser energy
- Electron beam
- Water jet
- Ultrasonic vibration

Types of Unconventional Machining

1. EDM (Electrical Discharge Machining)

Diagram

```
Electrode
    ↓
 Spark
```

```
 * * * *
____________
Workpiece
```

Working

- Sparks remove material.
- Used for electrically conductive materials.

Examples

- Dies
- Moulds
- Complex cavities

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2. ECM (Electrochemical Machining)

Diagram

```
Tool (-)
   ↓
Electrolyte
~~~~~~~~~~~~
Workpiece (+)
```

Working

- Material dissolves by electrolysis.
- No tool wear.

Examples

- Turbine blades
- Medical implants

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3. LBM (Laser Beam Machining)

Diagram

```
Laser Beam
    ↓↓↓
```

```
_____________
Workpiece
```

Working

- High-energy laser melts and vaporizes material.

Examples

- Micro holes
- Sheet cutting

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4. USM (Ultrasonic Machining)

Diagram

```
Tool
↓↓↓↓
Abrasive Slurry
~~~~~~~~~~~~~~~
Workpiece
```

Working

- Ultrasonic vibrations with abrasives remove material.

Examples

- Glass
- Ceramics
- Quartz

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5. AJM (Abrasive Jet Machining)

Diagram

```
Nozzle
 ↓↓↓↓↓
Abrasive Jet
_____________
Workpiece
```

Working

- High-speed abrasive particles erode the surface.

Examples

- Glass engraving
- Deburring

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6. WJM / AWJM (Water Jet / Abrasive Water Jet Machining)

Diagram

```
Nozzle
 ↓↓↓↓↓
Water Jet
__________

Workpiece
```

Working

- High-pressure water (with or without abrasives) cuts the material.

Examples

- Marble
- Rubber
- Aluminum
- Steel

Advantages of Unconventional Machining

- Machines very hard materials.
- Produces complex shapes.
- High accuracy.
- Minimal cutting force.
- Good surface finish.

Disadvantages

- High equipment cost.
- Lower material removal rate (for some processes).

- Requires skilled operation.
- Higher maintenance.

Difference Between Conventional and Unconventional Machining

Feature	Conventional Machining	Unconventional Machining
Tool Contact	Direct contact	Little or no direct contact
Material Removal	Mechanical cutting	Electrical, thermal, chemical, or jet energy
Tool Wear	High	Low or none (depends on process)
Cutting Force	High	Very low
Accuracy	Moderate	High
Surface Finish	Good	Excellent
Hard Materials	Difficult	Easy
Complex Shapes	Limited	Easily produced
Initial Cost	Low	High
Examples	Turning, Milling, Drilling, Grinding	EDM, ECM, LBM, USM, AJM, AWJM

Applications

Conventional Machining

- Automobile parts
- Machine shafts
- Gears
- Nuts and bolts
- Engine components

Unconventional Machining

- Aerospace turbine blades
- Medical implants
- Injection moulds and dies
- Electronics (micro-holes, PCBs)
- Glass and ceramic machining

Important Exam Questions

1. Define conventional machining with examples.
2. Define unconventional machining with examples.
3. Differentiate between conventional and unconventional machining.
4. Explain the working principle of EDM with a neat diagram.
5. Explain ECM with applications.
6. Explain Laser Beam Machining (LBM).
7. Explain Ultrasonic Machining (USM).
8. Explain Abrasive Water Jet Machining (AWJM).
9. List the advantages and disadvantages of conventional machining.
10. Compare EDM and ECM.
11. Compare AJM and AWJM.
12. Explain the applications of unconventional machining in aerospace and medical industries.